DNS Secondary Verifier

Frozen URL policy + DNS-only routed evidence • diagnostic 20K evaluation

System	Accuracy	Precision	Recall	F1	ROC-AUC	FPR	FNR
Primary only	97.55%	94.83%	79.85%	86.70%	97.52%	0.48%	20.15%
Primary + DNS	97.69%	92.30%	83.90%	87.90%	97.56%	0.78%	16.10%

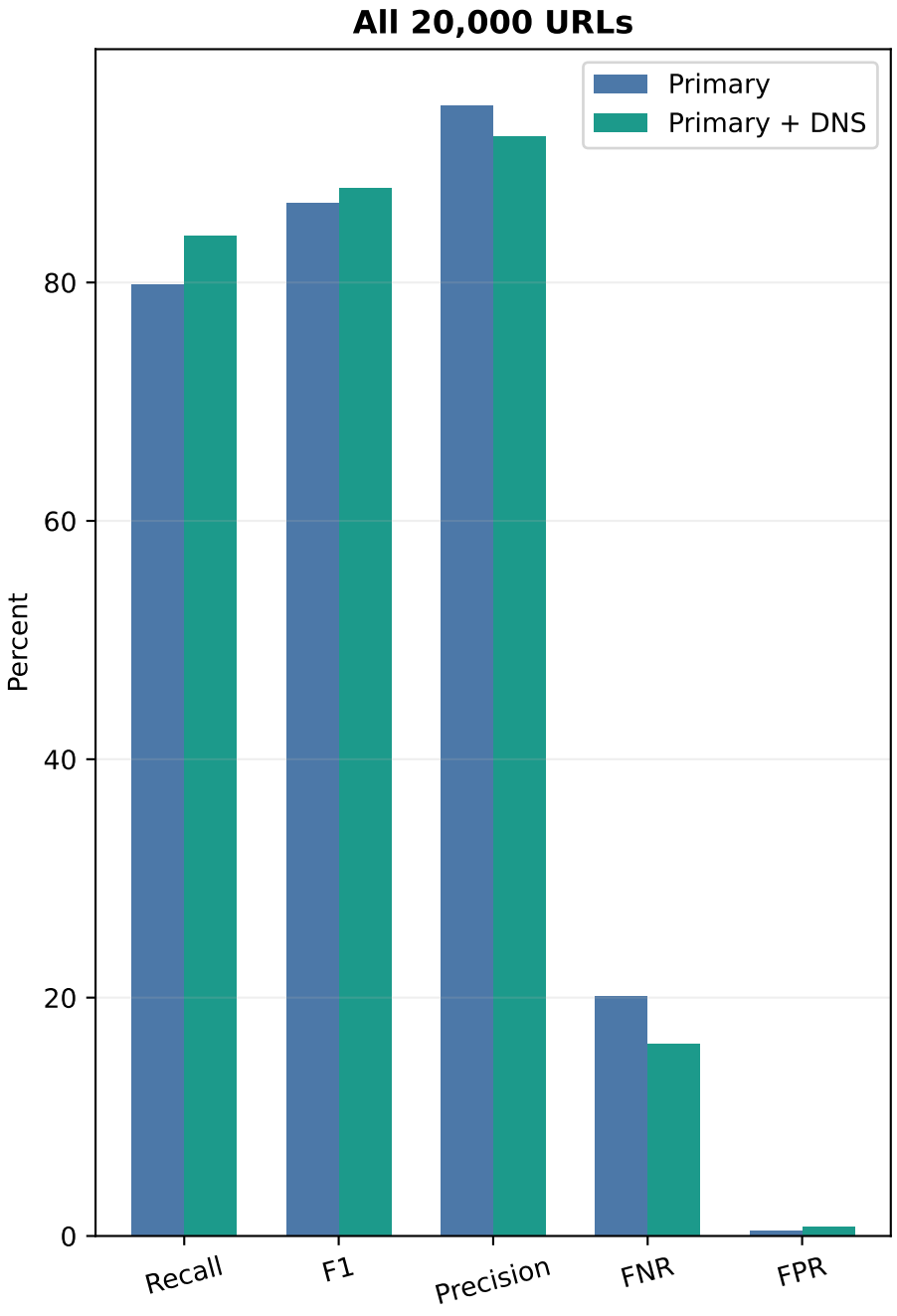
Recovered phishing: +81 **Additional false positives: +53**

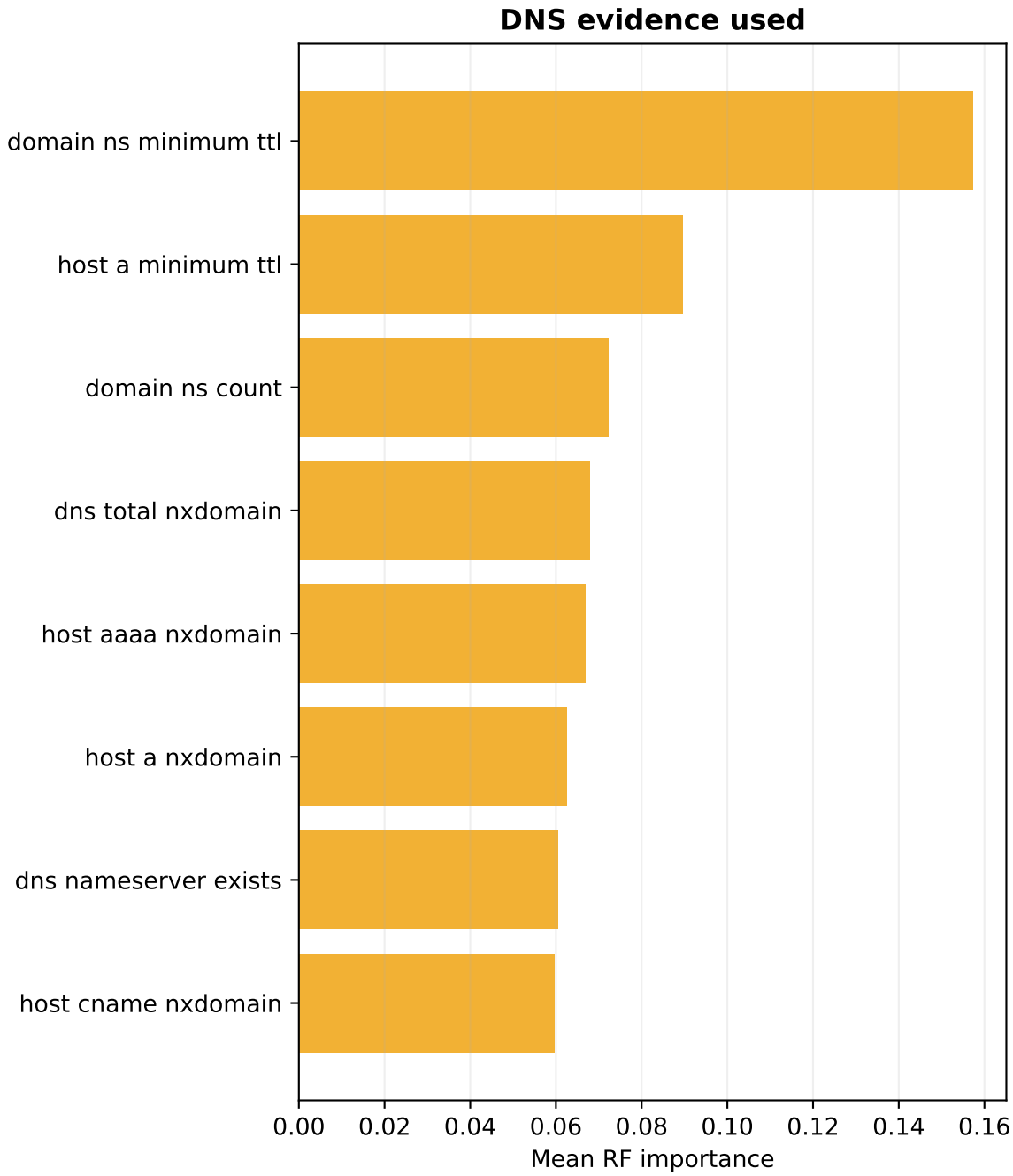
FNR change: -4.05 points **FPR change: +0.294 points**

The locked DNS branch recovered 81 additional phishing URLs and added 53 false positives. FNR changed from 20.15% to 16.10%; FPR changed from 0.48% to 0.78%. DNS helps modestly but does not solve the remaining URL-only misses.

The DNS model and threshold were locked on separate development data before these DNS records were collected. The 20K cohort was already viewed, so this remains diagnostic—not temporal confirmation.

Locked combined-policy comparison





Interpretation and Limits

Independent branch	The RF receives 25 DNS values only; it never sees URL characters, lexical values, primary scores, response time, or 20K labels.
Routed scope	Only 941 secondary-review URLs were queried. DNS ROC-AUC inside that difficult subset is 87.60%.
Latency	The concurrent offline collection took 53.0s for 941 domains. This is not acceptable evidence of browser latency.
Main risk	DNS absence and TTL can reflect takedown timing, resolver policy, or temporary failure. Their predictive value may decay rapidly.
Safety	No phishing page, redirect, TLS endpoint, WHOIS/RDAP service, or page content was opened.
Next proof	Repeat on a later low-prevalence snapshot with a cache and timeout policy; do not change the frozen model or threshold after seeing labels.